4 (a) (i) Using Fleming Left hand rule, the magnetic force will always be perpendicular to the direction of the charged particle's velocity and provides the centripetal force. Thus, it only changes the direction of velocity and not the speed. Therefore, the magnitude of the magnetic force will remain constant.

(ii) Negative.

(iii) Magnetic force provides the centripetal force,

$$Bqv = \frac{mv^2}{r}$$

$$q = \frac{mv}{Br}$$

$$q \propto \frac{1}{r} \quad \Rightarrow \quad \frac{q_x}{q_y} = \frac{r_y}{r_x} = 2$$

(b) (i) As particle X collides with the gas particles, its magnitude of velocity v will gradually decrease. Since $r = \frac{mv}{Bq}$ and only v is decreasing with B, charge q and mass m constant, r, the radius of circular motion, will be decreasing as depicted by the spiral nature in the diagram.

(ii) 1. $v = r\omega$, where $\omega = \frac{2\pi}{T}$ => $v = r\frac{2\pi}{T}$ => $T = 2\pi(\frac{r}{v})$

Hence, $T = 2\pi(\frac{m}{Bq})$ => $T = \frac{2\pi\, m}{B\, q_x}$

2. When $m = 3000(9.11\times10^{-31}) = 2.733 \times 10^{-27}$ kg, $B = 1.0$ T, $q = 1.6 \times 10^{-19}$ C, $T = \frac{2\pi\, m}{B\, q_x} = \frac{2\pi(2.733 \times 10^{-27})}{(1.0)(1.6 \times 10^{-19})} = 1.07 \times 10^{-7}$ s. It cannot be a tau particle as it does not live long enough to make the orbits shown in Fig. 4.2.

5 **(a)** **(i)** To channel / strengthen / increase / concentrate the magnetic field produced by